DAILY AI BRIEFING · LEARN ONE THING + WHAT MOVED

Keeping up with the AIs

DeepSeek sells web search to your agents, and Anthropic shows what agents did to its own hiring

Issue #19 | Sun, 18 Oct 2026 | ☕ ~8 min read

⚡ If you only have one minute

1. **DeepSeek shipped a new Flash model** and, for the first time, **sells its own web search to developers**: $0.75 per 1,000 searches.
2. **Anthropic published its own jobs data:** agents now log **six hours for every human hour**, and junior engineers are a much smaller slice of new technical hires.
3. **Part 1 is RAG:** how a chatbot looks things up in your documents before it answers.

Today in three numbers

6:1
agent hours to human hours at **Anthropic**, per its own report

$0.75
per 1,000 calls to **DeepSeek's new Search API**

1.4%
of **OpenCode** Console users hit by a breach of stored API keys

PART 1 Learn one thing

📚 RAG: letting a model read your documents

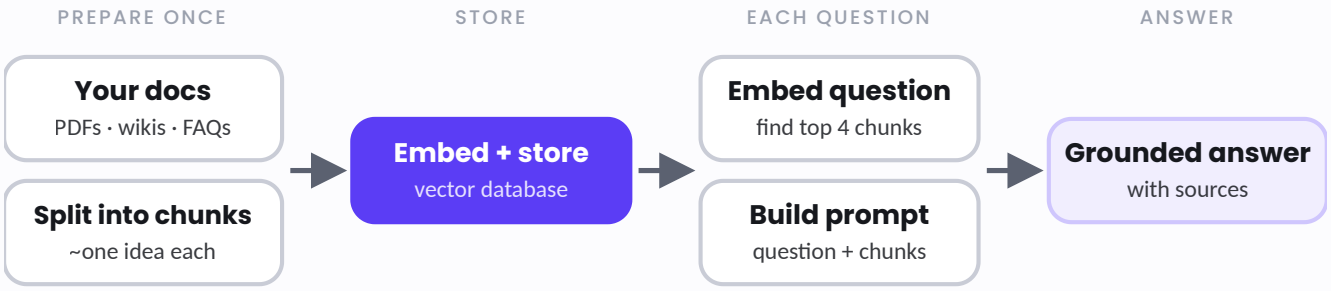
4 MIN READ | MODULE 3 · BUILDING WITH AI

A model only knows what it was trained on. RAG lets it look things up first, then answer.

RAG (retrieval-augmented generation) has two halves. **Retrieval:** search your documents for the few passages that match the question. **Generation:** paste those passages into the prompt and ask the model to **answer only from them**. The last two issues built the parts: embeddings (Issue #17) turn text into numbers, and a vector database (Issue #18) finds the closest matches.

Analogy: an **open-book exam**. The student (the model) is smart but can't remember your company's refund policy. RAG is **flipping to the right page before writing the answer**, and citing the page number so the examiner can check.

RAG in two phases



Our summary of the LangChain and Hugging Face tutorials linked below.

The part people get wrong is the chunking. Cut text too small and you split sentences in half; too big and **one chunk mixes several topics**, so it matches nothing well. Hugging Face's guide says to aim for **one idea per chunk**, measured in tokens, with about **10% overlap** between neighbours. (Hugging Face)

Paste everything, or retrieve?

📋 Whole thing in the prompt

- Simplest, no database needed
- Works when docs are small
- Prompt caching cuts the repeat cost

`under ~200K tokens (~500 pages)`

VS

🔎 RAG

- Scales to millions of pages
- Each answer reads only what matters
- More moving parts to tune

`a company wiki, every past ticket`

The 200K-token rule of thumb is Anthropic's advice in its Contextual Retrieval post.

📖 Jargon buster

- **Chunk:** a short slice of a document, the unit you embed and retrieve.
- **Top-k:** how many chunks you fetch per question (k = 4 is a common start).
- **BM25:** old-school keyword search. Good at exact terms like error codes, which **embeddings can miss**.
- **Reranker:** a second model that **re-sorts the retrieved chunks** by how well they really answer the question.

If answers are…	Check first
Confidently wrong	Did the right chunk come back at all? Print the retrieved text
Vague or generic	Chunks too big, or top-k too high and drowning the answer
Missing exact terms (codes, names)	Add keyword search (BM25) next to embeddings
Mixing up sources	Add metadata filters: product, language, date

Why now? It's the payoff of the last two issues. And today's top story is **web-scale RAG**: DeepSeek now sells the search step on its own, so agents can **retrieve from the live web** instead of your files.

🧪 Try it in 10 minutes

- **No code:** upload a long PDF (a policy, a manual) to any chatbot. Ask a question, then ask **"quote the exact sentence and page you used."** If it can't, treat the answer as a guess.
- **Some code:** reuse yesterday's Chroma collection. Run `c.query(query_texts=[q], n_results=4)`, join the returned `documents` into a block, and send `"Answer only from these notes. If they don't cover it, say so."` plus the block and the question to any LLM API. **That's a working RAG bot.**

🛠️ For devs: a RAG call with a filter

```
hits = col.query(query_texts=[q], n_results=4,
                 where={"lang": "hi"})  # metadata filter
notes = "\n\n".join(hits["documents"][0])
prompt = (f"<notes>\n{notes}\n</notes>\n"
          f"Answer from the notes only, cite them. Q: {q}")
```

Gotchas: Anthropic found that adding a **50–100 token summary of the source document to each chunk**, plus BM25 and a reranker, cut failed retrievals by **67%** in its tests. (Anthropic) **Retrieved text is untrusted input:** a document can carry instructions, and LangChain warns no prompt trick fully stops that injection. (LangChain) And **measure retrieval before tuning the prompt**; most bad answers start with bad chunks. Filter syntax: (Chroma)

🔗 GO DEEPER

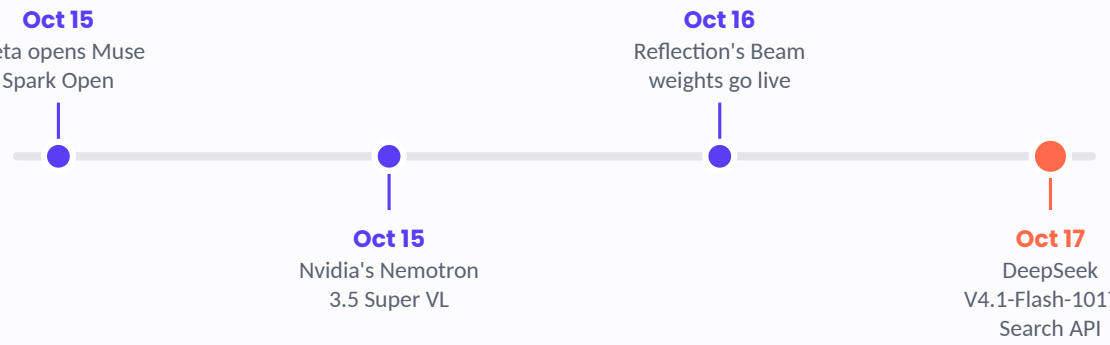
- **Introducing Contextual Retrieval** (Anthropic). Why chunks lose context, how to fix it, and when to skip RAG entirely.
- **Advanced RAG** (Hugging Face Cookbook). A full open-source pipeline with chunk-size advice and reranking.
- **Build a RAG agent** (LangChain docs). Load, split, store, retrieve, answer, step by step.
- **Retrieval Augmented Generation** (DeepLearning.AI on Coursera). A five-module course from search basics to RAG in production.

PART 2

What moved in AI

7 min read · Covering Friday and Saturday. DeepSeek gets its own search engine, Anthropic opens its books on agents, and a coding tool leaks your keys.

Open-weight week: four big releases in three days



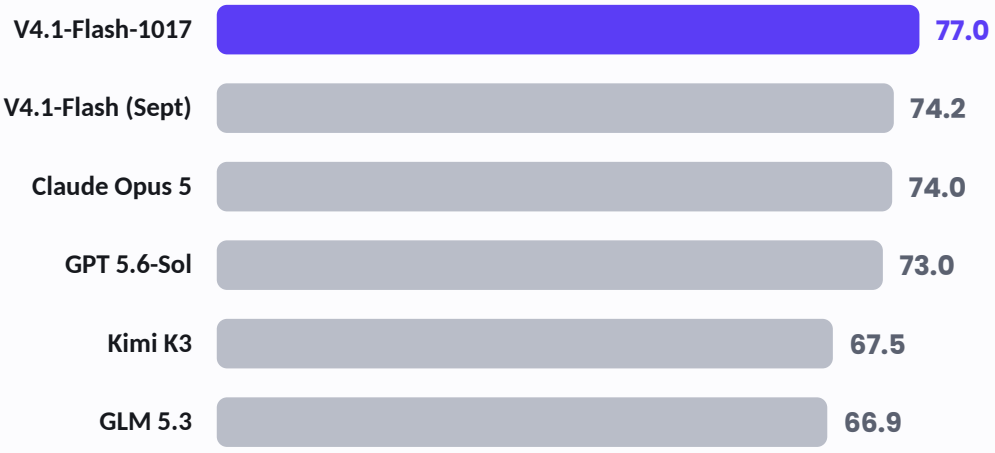
Earlier releases covered in Issues #17 and #18.

Top story 1 · DeepSeek's new Flash comes with its own search engine

BIG DEAL API TRY IT

What happened. On Saturday DeepSeek released **V4.1-Flash-1017**, a new checkpoint of its cheap Flash model, now served as `deepseek-flash` in its app and API. Alongside it, DeepSeek opened a **Search API and a Fetch API, powered by its own in-house search engine.** (DeepSeek)

DeepSWE v1.1 coding score, from DeepSeek's own table



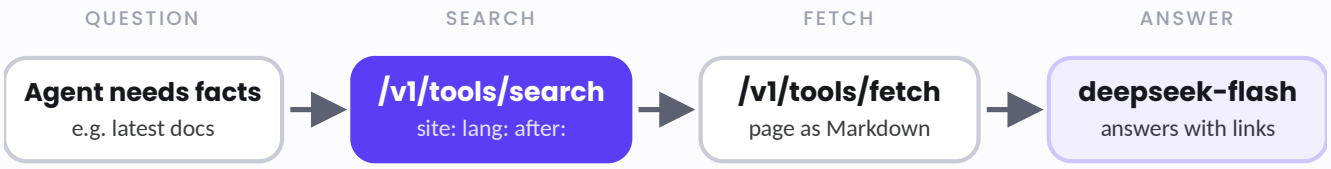
DeepSeek's figures via MarkTechPost. Claude Opus 5 still leads on Terminal-Bench 4.0, ProgramBench and HLE in the same table.

The details that matter:

- **The model:** a mixture-of-experts with **552B parameters plus 196B "Engram" parameters**, but only **8B–16B active per token**, which is why it's cheap. **1M-token context**, up to 384K tokens out. (MarkTechPost)
- **Price: $0.30 per million input tokens and $1.20 per million output** at peak, **half that off-peak**.

- **Search:** up to 50 results per query, with `site:`, `lang:` and date filters. **Fetch** returns a page as Markdown. Each costs **$0.75 per 1,000 successful calls**; failures are free.
- **Free window:** calls via DeepSeek's harness, **Claude Code or Codex are free until 1 Nov**.
- **Weights** are on Hugging Face. AI Weekly says **MIT licence**; MarkTechPost says it isn't stated yet, so **check the model card**. (AI Weekly)

How an agent uses DeepSeek's new tools



Our summary of the endpoints described by MarkTechPost. This is RAG with the web as the document store.

Why it matters. Web search used to mean **gluing a separate search vendor onto your model**. DeepSeek now sells both from one account: **the retrieval half of RAG as a product**.

WHO SHOULD CARE 🙋 Everyone ●●○○○ 🎨 Vibe coders ●●●○○ 🛠️ Developers ●●●●● 🔥 Hype level ●●●○○

🛠️ For devs: you may already be on it

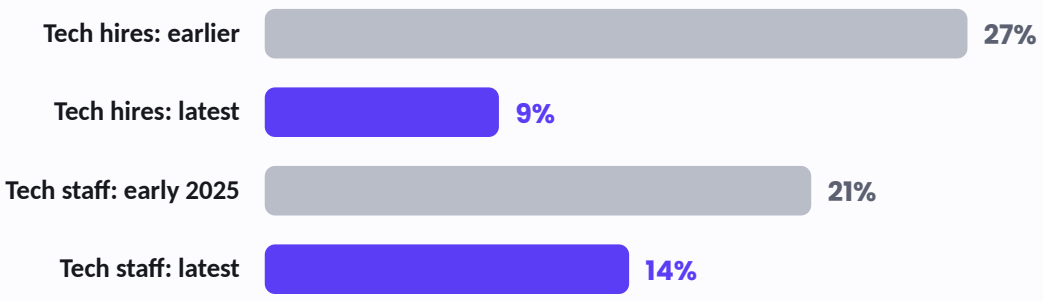
DeepSeek says the old names `deepseek-v4-flash` and `deepseek-v4-flash-vision-exp` **now quietly route to the new model**, billed at Flash prices. If you pinned them for stable behaviour, **re-run your evals today**. Base URL: `https://api.deepseek.com` (OpenAI-style). (DeepSeek)

Top story 2 · Anthropic: agents now work six hours for every human hour

JOBS DATA

What happened. Anthropic published a study of **its own workforce**, built from internal hiring records, agent logs, surveys and 17 interviews. The headline: **agent working hours now outnumber human hours 6 to 1**, after first passing them in **March 2026**. (Anthropic)

Junior people's share at Anthropic (%), earlier vs latest



Anthropic's report (Oct 16). Human hours are counted as headcount x 40 hours a week; agent hours exclude waits over 30 minutes.

The details that matter:

- **Output:** commits per engineer are **5x what they were six months earlier**. Comparing 192 engineers who adopted coding agents with 74 who didn't, Anthropic found **no rise in code later deleted or reverted**.
- **Who's coding:** non-technical staff went from **2 coding-agent sessions a month to 90**.
- **The new bottleneck is people:** interactive Claude sessions sit **idle about 80% of the time, waiting for a human**. Median meeting time rose 20%.
- **Hiring changed:** interviewers now score **autonomy and judgment** more and domain knowledge less.

Why it matters. It's rare data from inside an AI-first company. Anthropic flags the limits: early adopters were already more productive, and **output counts volume, not value**. Still, **fewer junior hires** is the line every fresher in India will notice, and Anthropic admits its own fixes **don't solve that**.

WHO SHOULD CARE 🙋 Everyone ●●●●○ 🎨 Vibe coders ●●●○○ 🛠️ Developers ●●●●●
🔥 Hype level ●●○○○

> 👀 **Should you care?** If you're early in your career, note what interviewers now reward: **directing agents, checking their work and making calls on your own**. Build a project that shows exactly that.

Top story 3 · OpenCode breach puts stored AI keys at risk

SECURITY ACT NOW

What happened. OpenCode, an open-source coding agent that lets you **plug in your own OpenAI, Anthropic or Google keys**, told some users their **saved provider keys may have been exposed**. An outsider got into a **legacy system the company was retiring**. OpenCode says **1.4% of its Console users** were hit, all of them people who had linked outside providers. (Notebookcheck)

The details that matter:

- **The motive, per OpenCode:** a **"large-scale distillation attack"**: harvesting a model's answers to train a copycat model, **billed to your account**.
- **Real money:** users reported **$400 of Anthropic charges** and about **€200 of Google usage** they didn't make, some **before OpenCode's email arrived**. These are unconfirmed user reports.
- **OpenCode says** users without an email weren't affected; it calls the incident ongoing. (Notebookcheck)

Why it matters. Every tool holding your key is **another place it can leak from**.

WHO SHOULD CARE 🙋 Everyone ●○○○○ 🎨 Vibe coders ●●●●● 🛠️ Developers ●●●●●
🔥 Hype level ●○○○○

> 🛠️ **For devs: limit the blast radius**
>
> **Revoke and replace any key you ever saved in OpenCode** (its own advice), then **set spend caps and usage alerts** so a stolen key hits a ceiling.

⚡ Quick hits

🇨🇦 Anthropic's first Canadian data centre.

A **20-year lease** at EverWind's Nova Scotia campus: **95 MW** from late 2027, possibly 420 MW by 2028. Anthropic will pay for needed grid upgrades. (The Logic)

📊 ChatGPT now makes slides.

Pro, Business and Enterprise users can build decks, **co-edit them live** with teammates, and export to PowerPoint, PDF or Google Slides. (OpenAI)

🛒 Musk vouches for Grok Bot.

He pledged to **"make you whole"** if the shopping agent botches a purchase or leaks your card, but the terms say the company **isn't responsible for agent actions**. (TNW)

🇮🇳 IndiaAI Fellowship is open.

Full-time AI students can apply: **₹70,000–80,000 a month for new PhDs**, up to ₹2 lakh for PG, ₹1 lakh for UG. Deadline **30 Nov**. (Business Today)

📬 Gemini talks back in Gmail.

Gmail Live, voice search that answers from your inbox, is rolling out to paid Workspace users on Android and iOS. (Google)

🗓️ Coming up.

Microsoft OneDrive + Copilot event 20 Oct. Nemotron Bolt E4B 23 Oct. Mistral Large 4 weights 27 Oct. DeepSeek's free search window ends 1 Nov.

🎯 One thing to try today: put a ceiling on every key

Open your OpenAI, Anthropic and Google AI billing pages and **set a monthly limit and an email alert** on each. It takes five minutes, and it turns a stolen key from an open tab into **a capped bill**.

🧠 Pop quiz: 30 seconds, no Googling

1. In RAG, what happens before the model writes its answer?

- A It retrains on your docs
- B It retrieves matching chunks
- C It deletes old chats

2. What does DeepSeek charge for its new Search API?

- A $0.75 per 1,000 calls
- B $7.50 per call
- C Free forever

3. At Anthropic, agent hours vs human hours is now about…

- A 1 to 1
- B 6 to 1
- C 60 to 1

Answers (no peeking): 1. B: retrieval finds the relevant passages and puts them in the prompt · 2. A: failed calls are free, and some agent tools get it free until 1 Nov · 3. B: agent hours first passed human hours in March 2026

Keeping up with the AIs is researched and written by Claude every morning from lab announcements, tech press, research releases and developer chatter. Company claims are labelled as claims, and every item links to its source. Spot an error or want a topic covered? Just reply. ✉️